

Problem-Solving and Programming Completed SPARK Reference

IslandCare Medical Centre - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: IslandCare Medical Centre

Sector: Medicine

Questions being answered

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the outstanding balance for each visit and the combined outstanding balance for all visits processed.

How SPARK completed it

- State what information the program receives.
- State the main calculation or decision the program performs.
- State the result the program must display.

Completed result

Problem statement: Calculate the outstanding balance for each visit and the combined outstanding balance for all visits processed.

2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

How SPARK completed it

- List inputs, processes and outputs before writing pseudocode.
- Use indentation to show the selection and loop structure.
- Keep variable names consistent from the algorithm through to the program.

Completed result

Inputs: Patient name, Age, Service code, Amount paid

Processes: Select service fee, Apply 5% senior discount at age 65 or over, Subtract amount paid, Accumulate balances

Outputs: Patient summary, Discount, Outstanding balance, Combined outstanding balance

Pseudocode:

START

SET totalBalance <- 0

INPUT visitCount

FOR visit <- 1 TO visitCount

INPUT patientName, age, serviceCode, amountPaid

SET serviceFee <- fee for serviceCode

IF age >= 65 THEN

SET discount <- serviceFee * 0.05

ELSE

SET discount <- 0

ENDIF

SET balance <- serviceFee - discount - amountPaid

SET totalBalance <- totalBalance + balance

DISPLAY patientName, discount, balance

NEXT visit

DISPLAY totalBalance

STOP

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

How SPARK completed it

- Choose values that exercise each branch of the selection statement.
- Follow the pseudocode in order and record important variable values.
- Calculate the expected output before running the program.

Completed result

The completed SPARK trace table includes the project's supplied normal examples and adds a boundary/validation discussion in the completed PDF.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as MedicalBalances_YourFullName.pas.

How SPARK completed it

- Create variables using suitable data types.
- Translate each pseudocode step into program statements in the same logical order.
- Compile or run often so errors are found in small sections.
- Use the same test data from the trace table.

Completed result

The SPARK reference implementation is MedicalBalances_YourFullName.pas. The completed PDF includes the full source code.

2. Test the program and capture evidence that the expected and actual results agree.

How SPARK completed it

- Run normal test data first.
- Run the boundary case.
- Run an invalid or unusual input when the program includes validation.
- Capture screenshots showing both the entered data and the program result.

Completed result

The completed SPARK reference compares expected trace-table outputs with the program output and records the result of each test.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

How SPARK completed it

- Create a cover page and table of contents.
- Keep the section order the same as the task sequence.
- Use readable code formatting and captions for screenshots.

- Export the final document to PDF and check every page.

Completed result

Completed documentation filename: ProgramDocumentation_YourFullName.pdf. The SPARK section PDF demonstrates the required order and evidence.

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Problem definition

Calculate the outstanding balance for each visit and the combined outstanding balance for all visits processed.

Input - Process - Output

Inputs	Processes	Outputs
Patient name Age Service code Amount paid	Select service fee Apply 5% senior discount at age 65 or over Subtract amount paid Accumulate balances	Patient summary Discount Outstanding balance Combined outstanding balance

Pseudocode

```
START
SET totalBalance <- 0
INPUT visitCount
FOR visit <- 1 TO visitCount
    INPUT patientName, age, serviceCode, amountPaid
    SET serviceFee <- fee for serviceCode
    IF age >= 65 THEN
        SET discount <- serviceFee * 0.05
    ELSE
        SET discount <- 0
    ENDIF
    SET balance <- serviceFee - discount - amountPaid
    SET totalBalance <- totalBalance + balance
    DISPLAY patientName, discount, balance
NEXT visit
DISPLAY totalBalance
STOP
```

Trace table and test data

Name	Age	Code	Fee	Paid	Discount	Balance
Marlon Reid	67	GEN	6500	3000	325	3175
Keisha Hall	41	WEL	8500	8500	0	0
Noel James	72	DIA	12000	5000	600	6400

Testing note

The normal data above should be supplemented with a boundary case and an invalid or unusual case where the program includes validation. Expected output should be calculated before the program is run.

Pascal reference implementation

```
program MedicalBalances;
var
  i, visitCount, age: integer;
  patientName, serviceCode: string;
  serviceFee, amountPaid, discount, balance, totalBalance: real;
begin
  totalBalance := 0;
  write('Number of visits: '); readln(visitCount);
  for i := 1 to visitCount do
    begin
      write('Patient name: '); readln(patientName);
      write('Age: '); readln(age);
      write('Service code: '); readln(serviceCode);
      write('Amount paid: '); readln(amountPaid);
      if serviceCode = 'GEN' then serviceFee := 6500
      else if serviceCode = 'WEL' then serviceFee := 8500
      else serviceFee := 12000;
      if age >= 65 then discount := serviceFee * 0.05 else discount := 0;
      balance := serviceFee - discount - amountPaid;
      totalBalance := totalBalance + balance;
      writeln(patientName, ' balance: ', balance:0:2);
    end;
  writeln('Combined balance: ', totalBalance:0:2);
end.
```

Documentation order

- Cover sheet
- Table of contents
- Problem definition
- Algorithm
- Source code
- Trace table and test data
- Program execution screenshots
- Final review

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

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